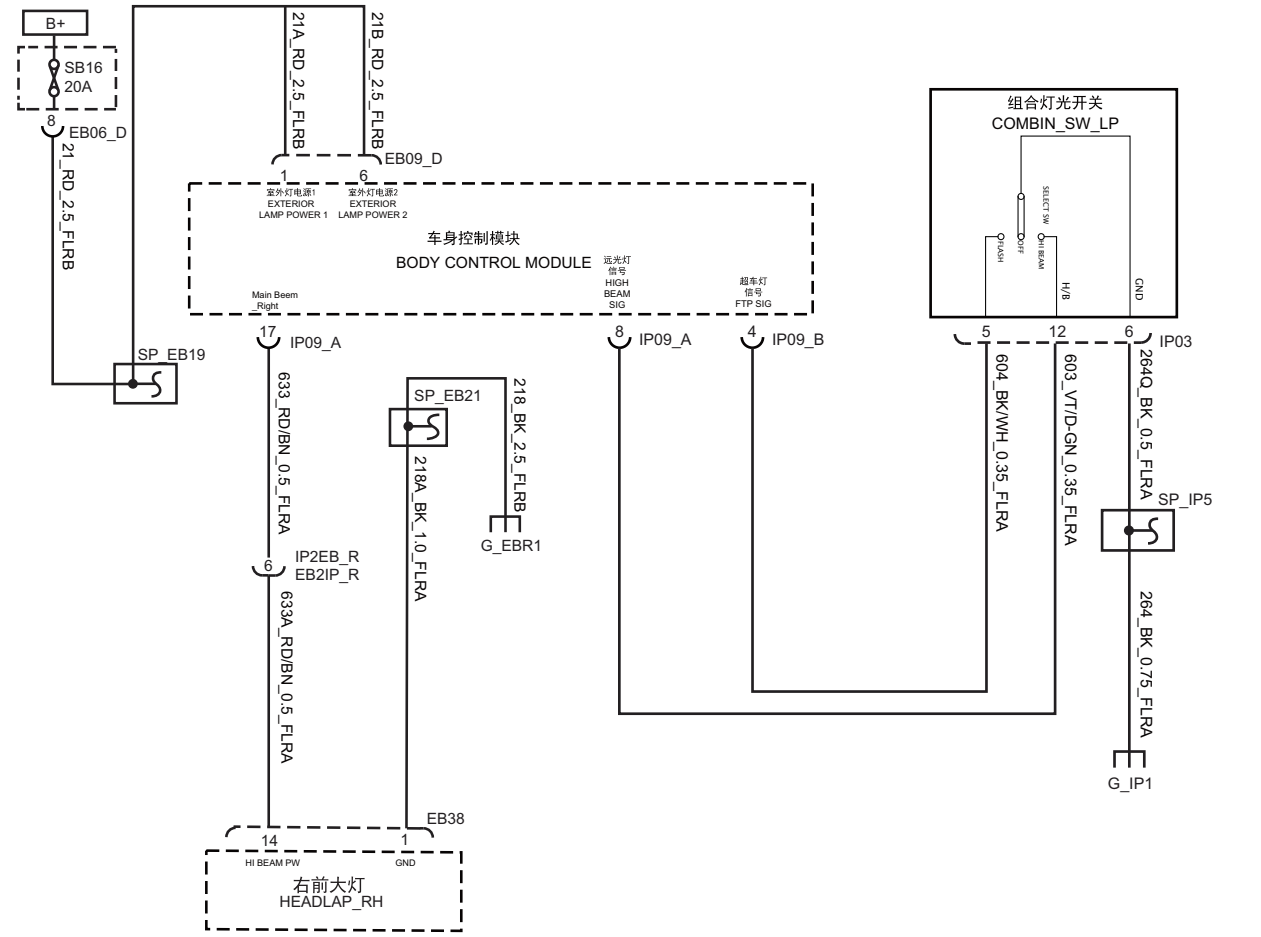
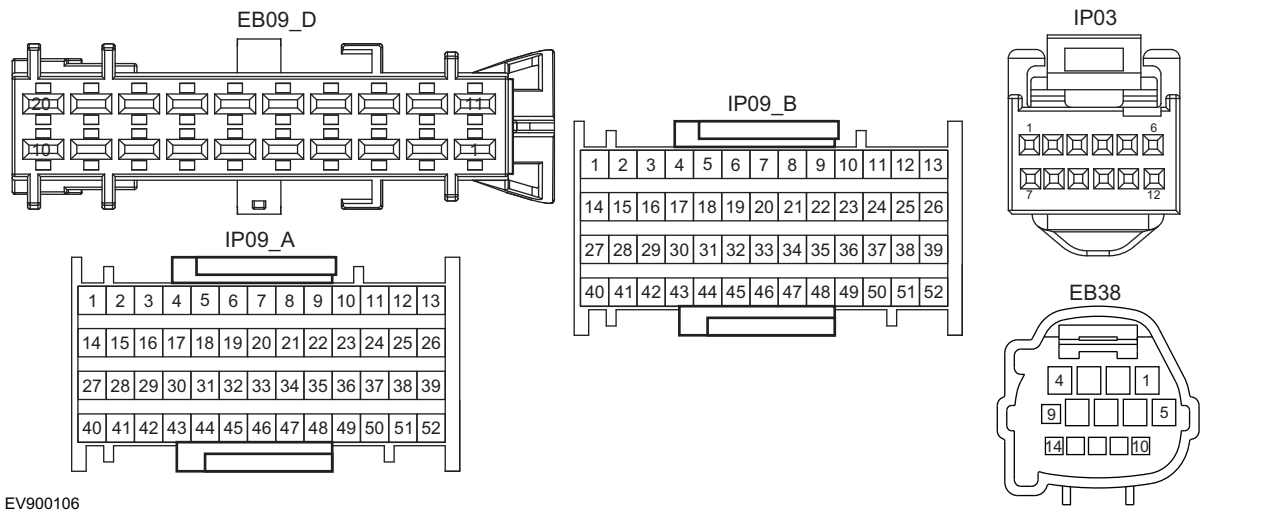
B1237-11、 B1237-12、 B1237-13

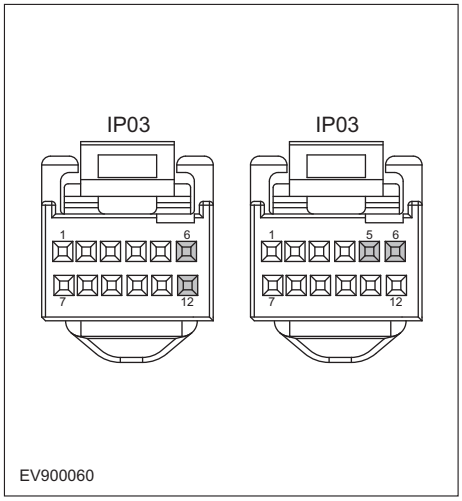
Fault diagnosis code
B1237-11：Right far light circuit short to ground
B1237-12：Right far light circuit short to battery
B1237-13：Right far light circuit open
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • BCM fault • Right high beam light circuit fault • Insurance fuse fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
B1237-11
BCM high beam light output circuit short to ground
B1237-12
BCM high beam light output circuit short to battery
B1237-13
BCM high beam light circuit open
To set the effect of a fault code condition
Dipped beam function failure
Description of circuit diagram

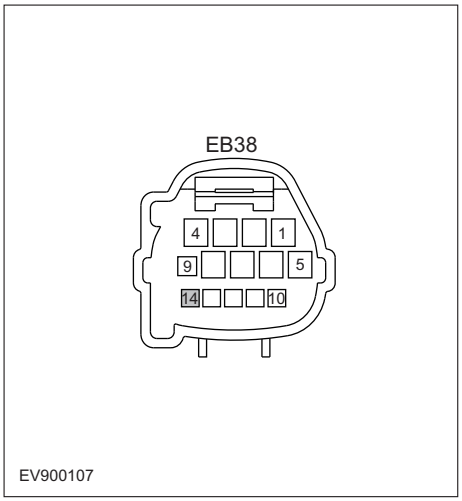
Circuit diagram

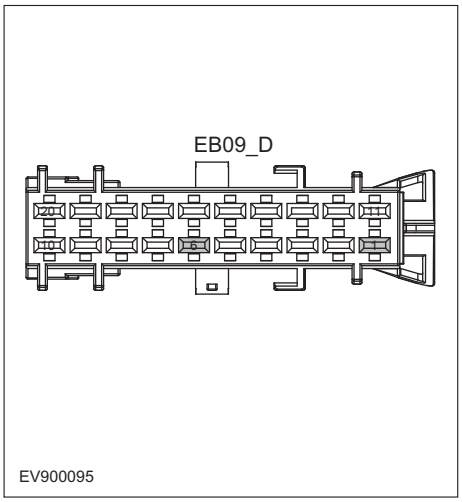


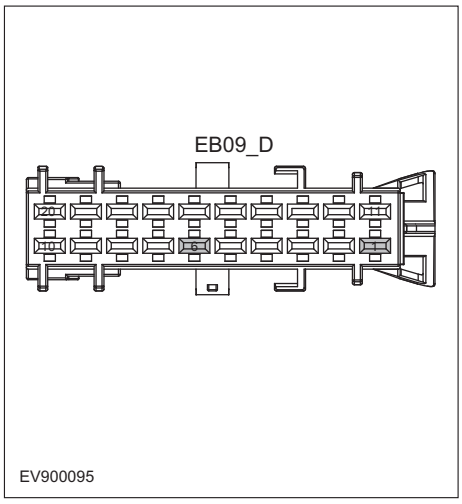
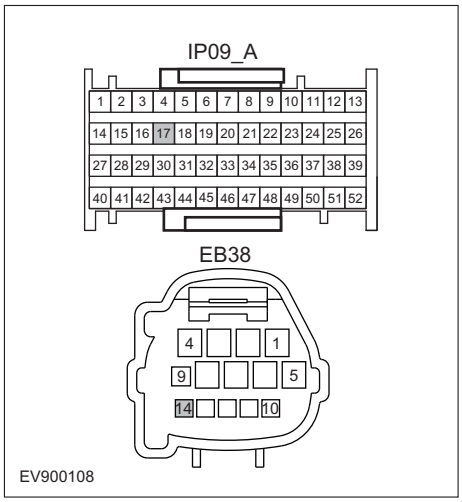


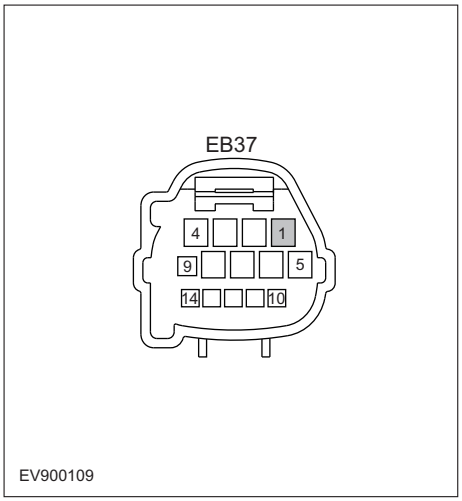
EV900106

<table>
<tr><td colspan="3">Troubleshooting steps</td></tr>
<tr><td>Step 1</td><td colspan="2">General inspection.</td></tr>
<tr><td colspan="2"></td><td>1 Check the fuse SB16 for damage.
2 Check battery capacity.
3 Check the harness plug of air conditioning control module for damage, poor contact, aging and looseness.
4 Check whether the result is normal or not?</td></tr>
<tr><td colspan="2"></td><td>NO ➞ Repair or replace the faulted parts.</td></tr>
<tr><td colspan="3">YES ↓</td></tr>
<tr><td>Step 2</td><td colspan="2">Check fault codes.</td></tr>
<tr><td colspan="2"></td><td>1 Connect the fault diagnosis instrument to the diagnosis interface.
2 Turn the ignition switch to ON position.
3 Check the body control module with a diagnostic apparatus.
4 Is there any other fault code except for B1237-11/B1237-12/B1237-13?</td></tr>
<tr><td colspan="2"></td><td>YES ➞ Refer to:DTC Summary list(BCM).</td></tr>
<tr><td colspan="3">NO ↓</td></tr>
<tr><td>Step 3</td><td colspan="2">Check the combination lights switch (high beam light switch)(components).</td></tr>
<tr><td colspan="2"> 
</td><td>1 Turn the ignition switch to OFF.
2 Disconnect the harness plug IP03 of IPK module.
3 Place the combination light switch in the “high beam light”position and measure the resistance between the combination light switch terminals(combination switch).
Measuring circuit: resistance between terminal 12 on plug IP03 and terminal 6 on plug IP03.
Standard value: ≈ 0Ω
4 Place the combination light switch in the “flashlight”position and measure the resistance between the combination light switch terminals(combination switch).
Measuring circuit: resistance between terminal 5 on plug IP03 and terminal 6 on plug IP03.
Standard value: ≈ 0Ω
5 Check whether the result is normal or not?</td></tr>
<tr><td colspan="2"></td><td>NO ➞ Replace combination lights.</td></tr>
<tr><td colspan="3">YES ↓</td></tr>
</table>

Step 4	Check the supply power of high beam light when it is started or operating.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB38 of right front high beam light. 3 Turn the ignition switch to ON. 4 Measure the voltage between the harness terminal of left front high beam light and ground. Measuring circuit: voltage between terminal 14 on plug EB38 and ground. **Standard value:10~14V** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 5	Check the power voltage of outdoor light of BCM
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB09_D of BCM. 3 Measure the voltage between the harness terminal of BCM and ground. Measuring circuit: voltage between terminal 1 on plug EB09_D and ground. Measuring circuit: voltage between terminal 6 on plug EB09_D and ground. **Standard value:10~14V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the power voltage of outdoor light when the BCM is started or operating
	1 Turn the ignition switch to ON. 2 Measure the voltage between the harness terminal of BCM and ground. Measuring circuit: voltage between terminal 1 on plug EB09_D and ground. Measuring circuit: voltage between terminal 6 on plug EB09_D and ground. **Standard value:10~14V** 3 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 7	Check the circuit between right front high beam light and BCM (open).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP09_A of BCM. 3 Measure the resistance between the harness terminal of right front high beam light and of BCM. Measuring circuit: resistance between terminal 14 on plug EB38 and terminal 17 on plug IP09_A. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 8	Test whether the grounding circuit of high beam light is open.
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness terminal of right front high beam light and ground. Measuring circuit: resistance between terminal 1 on plug EB38 and ground. **Standard value: < 1Ω** 3 Test whether the resistance is less than 1Ω?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 9	Test whether BCM is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB09_C、EB09_D、IP09_A、IP09_B of BCM. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BCM plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the BCM.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	
Next Step ↓	
Diagnosis end.	